

SigDINO-Qwen-GateX: A Frozen-Encoder Multimodal Fusion Pipeline

for Fake Review Detection — Training Report

October 1, 2026

Machine: NVIDIA GeForce RTX 3050 Laptop GPU (6 GB VRAM), CUDA 12.1
Project folder: C:\FAKE REVIEW

1 Architecture

Encoders (all frozen, no fine-tuning):

- Text: `answerdotai/ModernBERT-base` — token-level sequence output
- Image A: `google/siglip2-base-patch16-224` — patch tokens plus image/text embeddings used for a cosine alignment score
- Image B: `facebook/dinov2-base` — patch tokens, fine-grained

Fusion head (the only trainable component, 3,875,842 parameters):

$$T_{\text{seq}} = \text{proj}_{\text{text}}(\text{ModernBERT token sequence}) \in R^{B \times L_t \times D} \quad (1)$$

$$V_{\text{img_seq}} = \text{concat}(\text{proj}_{\text{siglip}}(\text{SigLIP2 patches}), \text{proj}_{\text{dino}}(\text{DINOv2 patches})) \in R^{B \times N_{\text{img}} \times D} \quad (2)$$

$$H_{\text{text}} = \text{meanpool}(\text{CrossAttn}(Q=T_{\text{seq}}, K=V=V_{\text{img_seq}})) \quad (3)$$

$$H_{\text{img}} = \text{meanpool}(\text{CrossAttn}(Q=V_{\text{img_seq}}, K=V=T_{\text{seq}})) \quad (4)$$

$$g = \sigma(W_g [H_{\text{text}}; H_{\text{img}}; s_{\text{align}}]) \quad (\text{gated multimodal unit}) \quad (5)$$

$$F = g \odot H_{\text{text}} + (1 - g) \odot H_{\text{img}} \quad (6)$$

$$\text{logits} = \text{MLP}([F; s_{\text{align}}]) \quad (7)$$

This is a two-stage pipeline designed to fit a 6 GB laptop GPU:

Stage 1 (`extract_features.py`): runs the three frozen encoders once, one at a time, with no gradients, and caches their outputs to disk.

Stage 2 (`train.py`): trains only the small fusion head on the cached tensors; the large encoders are never loaded in this stage.

2 Dataset

AiGen-FoodReview (Gambetti & Han, ICWSM 2024), Zenodo record 10511456, MIT licensed. Review text and restaurant image pairs, binary label (human-written vs. AI-generated review/image pair).

Quantity	Value
Total pairs downloaded	20,144 images + 3 CSVs (train/val/test)
Train split	12,086 samples
Validation split	4,028 samples
Test split	4,030 samples
ID column in CSVs	ID (uppercase)

Table 1: Dataset split sizes.

3 Bug Found and Fixed During This Run

File: `extract_features.py` (SigLIP2 stream).

Problem: in the installed `transformers` version, `model.get_image_features()` and `model.get_text_features()` return a `BaseModelOutputWithPooling` object rather than a raw tensor, as the original code assumed. This crashed the smoke test with:

```
AttributeError: 'BaseModelOutputWithPooling' object has no attribute 'float'
```

Fix: verified live against the actual loaded SigLIP2 model, then changed both calls to read `.pooler_output` from the returned object. The smoke test (50 samples) was re-run end to end afterward to confirm the fix; it passed cleanly with the expected tensor shapes.

4 Feature Extraction (Stage 1) — Result

Ran successfully on GPU for all three splits with no errors.

File	Samples	Size
<code>features/train.pt</code>	12,086	2.97 GB
<code>features/val.pt</code>	4,028	0.99 GB
<code>features/test.pt</code>	4,030	0.99 GB

Table 2: Cached feature files.

Cached tensors per sample: `text_tokens` (128, 768, ModernBERT, fp16), `text_mask` (128, bool), `siglip_patches` (16, 768, pooled 4×4 grid, fp16), `dino_patches` (16, 768, pooled 4×4 grid, fp16), `s_align` (1, SigLIP2 image-text cosine alignment score), `label` (0 = human, 1 = AI-generated).

5 Training (Stage 2) — Result

Command: `python train.py --features features --out runs/exp1`

Trainable parameters: 3,875,842 **Batch size:** 64 **LR:** 2×10^{-4} **Max epochs:** 20

Early-stop patience: 5

6 Final Test Set Result

Evaluated on the held-out 4,030-sample test split using the best (epoch 7) checkpoint.

Epoch	Train Loss	Val Loss	Val Acc	Val F1	Note
1	0.0917	0.0253	0.9918	0.9917	new best
2	0.0084	0.0096	0.9960	0.9960	new best
3	0.0049	0.0590	0.9856	0.9854	
4	0.0018	0.0102	0.9973	0.9973	new best
5	0.0040	0.0097	0.9970	0.9970	
6	0.0045	0.0105	0.9970	0.9970	
7	0.0010	0.0048	0.9985	0.9985	new best (checkpoint used)
8	0.0015	0.0121	0.9975	0.9975	
9	0.0011	0.0051	0.9978	0.9978	
10	0.0002	0.0199	0.9948	0.9948	
11	0.0000	0.0073	0.9978	0.9978	
12	0.0000	0.0072	0.9983	0.9983	

Table 3: Per-epoch validation metrics. Training early-stopped after epoch 12 (no val F1 improvement for 5 epochs). Best checkpoint: epoch 7 (val F1 = 0.9985), saved as `runs/exp1/best_model.pt`.

Metric	Value
Loss	0.0034
Accuracy	99.93%
Precision	99.85%
Recall	100.00%
F1 score	99.93%

Table 4: Final test-set metrics (`runs/exp1/test_metrics.json`).

7 Ablation Study: Text-Only and Image-Only Baselines

A test accuracy of 99.93% is unusually high for a fake-review detection task, so the two ablations recommended by the architecture document and the README were run: **text-only** (image stream and alignment score zeroed) and **image-only** (text stream and alignment score zeroed), using `train.py --ablation {text_only,image_only}` against the same cached features and the same fusion-head architecture and training recipe.

Model	Test Acc	Test Prec	Test Recall	Test F1	Epochs to best
Full fusion	99.93%	99.85%	100.00%	99.93%	7 of 12 run
Text-only	99.83%	99.90%	99.76%	99.83%	6 of 11 run
Image-only	99.21%	99.03%	99.41%	99.22%	18 of 20 run

Table 5: Ablation comparison on the held-out test split.

Conclusions:

- Text alone already reaches very close to the full model (99.83% vs. 99.93% F1) — text is the dominant signal for distinguishing human-written from AI-generated reviews in AiGen-FoodReview, consistent with it being reported as an “easy” benchmark in the literature.

- Image alone is noticeably weaker (99.22% F1) and required the full 20-epoch budget to reach it, confirming the image stream carries a real but smaller and harder-to-extract signal than text.
- The full-fusion result is *not* simply “the easy modality alone”: cross-attention fusion recovers the remaining ~ 0.1 -point gap on top of the text-only baseline, aided by the SigLIP2 image–text alignment score, the gated combination, and the dual-stream patch features. The improvement is small in absolute terms because the dataset is close to saturated by text alone, but it is a real, measured effect.
- None of this suggests a data leak: train, validation, and test sets come from disjoint CSVs, and features were extracted independently per split.

Engineering note: the first text-only attempt crashed with a Windows-specific `DataLoader` multiprocessing error (`RuntimeError: Couldn't open shared file mapping...`, error code 1455 — a page-file/shared-memory limit, unrelated to the model code). It did not reproduce on retry with `--num-workers 0` (single-process data loading), which is what the reported text-only numbers above use.

8 Simplifications Made (for the Writeup)

- Text encoder is ModernBERT-base, not Qwen3-Embedding (Qwen3-Embedding is pooled-only and does not suit token-level cross-attention as-is).
- Image patch tokens are pooled to a 4×4 grid (16 tokens per stream) to keep cross-attention and the on-disk cache laptop-sized; `--pool-grid` can be raised later for a higher-resolution final run.
- All three encoders are kept frozen; no LoRA fine-tuning is implemented in this version.

9 Files Produced

```
extract_features.py (bug-fixed, verified)
train.py
download_dataset.py
requirements.txt
README.md
data/ (dataset: CSVs + 20,144 images)
features/{train,val,test}.pt (cached encoder outputs, ~4.9 GB total)
runs/exp1/best_model.pt (best fusion-head checkpoint)
runs/exp1/history.json (full per-epoch metrics)
runs/exp1/test_metrics.json (final test metrics)
runs/ablation_text_only/{best_model.pt,history.json,test_metrics.json}
runs/ablation_image_only/{best_model.pt,history.json,test_metrics.json}
train.py (now supports --ablation none|text_only|image_only)
```

10 Suggested Next Steps

1. **[Done]** Text-only and image-only ablations — see Section 7.
2. Consider `--pool-grid 7` (or higher) for more spatial detail on the image streams, since image-only is the weaker modality and more resolution may close some of the gap.

3. Consider LoRA-adapting the encoders (currently fully frozen) as a further refinement once the above is tried.
4. For a stronger paper/thesis claim, also compare against a cheaper non-fusion baseline (e.g. plain concatenation instead of cross-attention + GMU) to show the specific fusion mechanism, not just multimodality in general, is responsible for the gain over the text-only baseline.